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THE EFFECT OF GEOMETRY ON THE AERODYNAMICS OF A PROTOTYPE GAS TURBINE COMBUSTOR

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ABSTRACT

The method of admission of the swirling flow to the combustion chamber has a strong influence on the flow field structure in Gas Turbine Combustors (GTC). Two different exit configurations are studied. The first configuration is that of a swirl cup that ends only with a splash plate such that there is a sudden unguided expansion as the flow emanates from the swirl cup. The second is a swirl cup that ends with a splash plate and an asymmetric combustion dome. Laser Doppler Velocimetry (LDV) measurements are conducted in the horizontal plane (X-Y), for both configurations, 5mm from the flare exit. Also, LDV measurements are conducted in two vertical planes passing by the combustor centerline (X-Z and Y-Z). The results reveal a significant difference in the flow structure for both configurations. The combustion dome appears to reduce the turbulence activities close to the exit of the swirl cup. In addition, the presence of the combustion dome eliminates the corner recirculation zone and the low velocity region close to the combustor walls. It is interesting to see that the asymmetry of the combustion dome (9° difference in the expansion angle on both sides) results in a significant asymmetry in the velocity magnitude as well as the turbulence activities. Moreover, the asymmetry in the combustion dome results in a tilting of the CRZ toward the surface with the higher expansion angle. The results highlight the importance of the proper and careful design of the GTC front section. The experiments are conducted in a dump combustor (rectangular cross section). To study the effect of the chamber geometry on the flow field, the base configuration is installed in an annular combustor sector and LDV measurements are conducted in the axial radial plane (X-Z). The flow field as well as the shape of the CRZ are significantly different in both cases. The CRZ height reduced by 40% with the swirl cup installed to the SAC sector. The results emphasize the strong influence of the confinement on the flow structure.

1 INTRODUCTION

The requirements of low emissions and stable combustion are among the first priorities in the development and design of Gas Turbine Combustors (GTC). Aerodynamics play a vital role in stability, emissions as well as dynamics issues related to GTC. Researchers have been studying, for decades, aerodynamics in reacting and non-reacting flow fields. Significance of the aerodynamics in the combustion process is evident even in a simple process like igniting a cigarette lighter or even a match.

Swirling flow is commonly utilized in GTC. The idea depends on the generation of a Central Recirculation Zone (CRZ) which, as the name implies, recirculates a portion of the hot gases back to ignite the fresh coming mixture. The flow structure, factors affecting the size of the CRZ and different methods of swirl generation for simple swirlers are well documented (for example, ref [1-3]).

The CRZ has a highly periodic asymmetric instantaneous structure (ref [4-10]). The research conducted by Sadanandan, R. [11] showed that the small scale instantaneous eddies that develop in the flow field have very strong influence on the orientation of the reaction zone.

Few years ago, the swirl cup was introduced as a unique device to prepare the air fuel mixture. Many studies were conducted on counter-rotating swirl cup. The flow field has been studied by Mehta et al. [12]. The empirical design approaches as well as the Computational Fluid Dynamics (CFD) were conducted by Mongia et al. [13-19]. The Reynolds number effect on the flow structure of a counter rotating swirl cup was studied by Yongqiang Fu et al. [20]. The effect of fuel and equivalence ratio on spray was also studied by Yongqiang

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Fu et al. [21]. In addition, Yongqiang Fu, et al. studied the Confinement effect on the flow field [22]. Mongia [23] discussed in great detail, the benchmark experiments that were conducted on the swirl cup and compared the measurements to the CFD results. Unfortunately, according to Mongai [23], there is still a gap between CFD and measurements.

2 NOMENCLATURE

Va	=	Axial velocity component
Vr	=	Radial velocity component
Vt	=	Tangential velocity component
RMSA	=	Root mean square value of the axial velocity
RMSR	=	Root mean square value of the radial velocity
RMST	=	Root mean square value of the tangential velocity
R	=	Flare exit radius
CFD	=	Computational Fluid Dynamics
SAC	=	Single Annular Combustor
PIV	=	Particle Image Velocimetry
LDV	=	Laser Doppler Velocimetry
GTC	=	Gas Turbine Combustor
CRZ	=	Central Recirculation Zone
DPSS	=	Diode Pumped Solid State
X, Y, Z	=	Cartesian coordinate system with the origin defined at the flare center

3 EXPERIMENTAL SETUP AND TEST CONDITIONS:

3.1 Swirl cup and test sections:

The counter rotating coaxial radial swirlecup arrangement used in this study is illustrated in Figure 1. The inner swirler is connected to a venturi tube as depicted in Figure 1. The area ratio of the secondary to primary swirler is greater than 1. The secondary swirler is connected to a conical flare. The effective area of the swirl cup is 0.62 in^2. Two exit configurations are being tested. We refer to them as the base configuration and configuration 1. The base configuration has a splash plate and a combustion dome attached to the exit as depicted in Figure 2 The dome expansion angle is not symmetric. There is a difference of 9 degrees in the expansion angle of both sides of the dome. Configuration 1 is also shown in Figure 2. It is only equipped with a splash plate. The hardware is tested within a rectangular plexi-glass chamber (dump combustor). The dimensions are 133.6 x 114.3 x 330.2 mm respectively. The thickness of the plexi-glass test section is 1.6 mm. In order to study the confinement effect, the base configuration is also installed in a Single Annular Combustor (SAC) sector. All the SAC dilution holes are blocked to limit the study to the effect of the geometry on the flow field. The boundaries of the SAC sector are shown in Figures 25 and 26. The sector has a trapezoidal cross section (it is a sector of an engine). The width to breadth ratio of the SAC sector is the same width to breadth ratio of the rectangular chamber (85%). The SAC sector ends with a contraction toward the exit (usually combustor are designed in a way to enable attachment to the turbine nozzle ring). However, the geometrical details of the SAC are discussed elsewhere [24].

3.2 Experimental facility:

A schematic of the facility utilized is presented in Figure 3. A 203 mm PVC pipe is used as a manifold. A honeycomb inside the manifold provides uniform flow to the test section. Pressure sensor and thermocouples are mounted inside the manifold for monitoring of pressure and temperature at the test section inlet. Air enters the manifold through a 50 mm pipe which can provide pressurized air up to 1 kg/s at 6.8 bars. An atomizer generating olive oil seeding particles (1~5 μm) is connected to the manifold through a seeding port. A blower and an exhaust system are mounted above the test section to collect aerosols. To enable testing in the rectangular plexi-glass chamber, the swirl cup is mounted on a flange attached to the manifold and the plexi-glass chamber is installed on the flange. Later, the swirl cup is assembled to the SAC sector and measurements are repeated.

3.3 Diagnostics:

The two component LDV measurements are conducted using an Artium Technologies Inc. system. The system uses two diode-pumped solid-state (DPSS) lasers as the light source which doesn't require air or water cooling. The system comprises a transmitter, a receiver, two ASA signal processors and a computer. The 500 mm focal length transmitter and 300 mm focal length receiver are mounted on a computer controlled traverse.

3.4 Test conditions, data acquisition, measurement grid and uncertainty:

Experiments are conducted for isothermal flow conditions under ambient pressure and a temperature of 70±1ºF. LDV measurements are conducted at a total pressure drop (or total pressure loss) of 4.9%.

The data count is typically more than 7000 for the green laser and 3000 for the blue laser. The uncertainty in the mean velocity is within 0.1-0.3 m/s.

The measurements coordinate system is defined such that the axial direction of the swirler is designated as the Z direction with the zero set at the flare exit. The X direction is perpendicular to the Z and in the expanding dome plane. The Y direction is perpendicular to both Z and X. The coordinate system for both configurations under investigation is illustrated in Figure 4.

Measurements are conducted in the axial radial plan (X-Z), the tangential radial plan (Y-Z) plane as well as the horizontal plane (X-Y). Both vertical measurements planes pass by the swirler centerline.

The horizontal measurement plane is located 5mm from the flare. The measurement grid has a resolution of 2mm in both the X and Y directions. The vertical plane measurements are conducted at axial locations (Z/R) of 0.2, 0.4, 0.8, 1.6, 2, 4, 6, 6.68 and 8. The measurement grid has a fine resolution of 1mm in both the radial and tangential directions.

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The SAC sector vertical plane measurements are conducted with a resolution of 2mm in both the axial and radial direction.

4 RESULTS AND DISCUSSIONS

4.1 Horizontal plane (X-Y) measurements for both configurations:

The axial velocity contours for configuration 1 and the base configuration are presented in Figure 5 and 6 respectively. The results are non-dimensionalized by the flare radius (R). There are two black contours. The inner one shows the edge of the CRZ (contour of zero axial velocity). The outer one shows the edge of the flare. The axial velocity magnitude within the CRZ is typically 10 m/s. Outside of the CRZ, we notice high velocities indicating the presence of the annular high speed jet emanating from the swirl cup.

The effect of the splash plate edge on the flow field is considerable. Figure 5 shows that the flow field is axi-symmetric only in the central region and up to the axial velocity contour of 25 m/s. The region outside the 25 m/s contour is not axi-symmetric because of the presence of the splash plate lip (see Figure 4). The lip of the splash plate has a height of 5mm yet, it affects the velocity magnitude significantly as depicted in Figure 5. The velocity in the bounded portion of the splash plate (X-Z plane) reaches roughly 20 m/s. However, the unbounded region (Y-Z plane) reaches velocities up to 30 m/s.

The dome effect on the flow field is apparent as shown in Figure 6. The asymmetry in the axial velocity magnitude is attributed to the gradual expansion in the X-Z plane compared to the abrupt expansion in the Y-Z plane. In addition, we notice that the velocities are higher along the +ve X-direction than the -ve X direction. This is due to the slightly higher expansion on the +ve X direction. This effect will be further illustrated when the measurements in the axial radial plane are presented.

The contours of the root mean square (rms) value of the axial velocity (RMSA) in the horizontal plane for configuration 1 and the base configuration are presented in Figures 7 and 8 respectively. It is of prime importance to delineate regions with high turbulence for they become candidates for the development of periodic oscillations. The velocity oscillations in the vicinity of the flame zone strongly influence the flame response. The core of the CRZ is a region of low turbulence fluctuations as illustrated in Figures 7 and 8. However, the edge of the CRZ is a region of high turbulence activities. The turbulence activities are maximized in the annular jet region. Also, both figures show that the turbulence activity is relatively higher on the unbounded regions than the bounded regions. We already showed that the velocities follow the same trend. The presence of the dome damps the oscillations as depicted in Figure 8. In addition, the dome reduces the sensitivity of the flow to induced oscillations by eliminating the Outer Recircualtion Zone (also known as corner recirculation zone). This highlights the importance of gradually expanding the swirling flow to achieve higher flame stability.

4.2 Vertical plane (X-Z) measurements for both configurations:

4.2.1 Axial velocity and turbulent fluctuating component:

The axial velocity profiles for configuration 1 and the base configuration are presented in Figure 9 and 10 respectively. Figure 9 shows typical double humped axial velocity profiles near the swirler exit. As the flow proceeds downstream, the humps move away from the centerline indicating the growth of the CRZ (profiles from Z/R=0.2 to 1.6). A point is reached at which the maximum velocity occurs close to the dump combustor wall (Z/R=2). Further downstream, the CRZ starts closing and it eventually disappears at Z/R=6. A better representation of the results is obtained by plotting the contours of axial velocities as presented in Figure 11 (configuration 1). The contour plot shows that the flow field is fairly symmetric. This is also illustrated by plotting the streamlines in Figure 11. The black contour represents the edge of the CRZ. We observe that the CRZ extends up to 6.5R. Only the flow field near the swirl cup exit (roughly up to 3R) is of importance (roughly the location of the primary jets). In a real combustor, this is what we refer to as the primary zone or the dome region. In this region the swirling flow is the dominant flow. The CRZ reaches a maximum width of 2R at a downstream distance of 1.8R.

Figure 10 shows the axial velocity profiles for the base configuration. The effect of the dome on the flow field is enormous. We noticed that the profiles show no double humps along the expanding surfaces of the combustion dome. In other words, the combustion dome eliminates the regions of low velocity close to the combustor wall. Also, the velocity profiles show that the combustion dome eliminates the corner recirculation zone (compare Figures 11 and 12).

Figure 12 shows the contours of the axial velocities for the base configuration in the X-Z plane. The contour plot shows that the flow field is not symmetric. The velocities on the right hand side of the chamber are relatively higher than the left hand side. Recall that the right hand side is the one with higher expansion angle. In other words, the asymmetry of the dome plays a major rule in the asymmetry of the flow field. The CRZ appears to be tilted toward the left hand side as shown in Figure 12. It is really interesting that only 9° difference results in such impact. It is even more interesting that the effect is convected downstream up to the end of the combustor (see the velocity profile at Z/R=8).

The RMSA contours in the X-Z plan for configuration 1 are presented in Figure 13. The turbulence activities are high close to the swirler exit at the expanding annular jet region. Also, the turbulence activities are high in vicinity of the edge of the CRZ. There is no doubt that the edge of the CRZ is a very critical region as the flow is switching direction. However, the turbulence decays downstream as the flow becomes fully developed. The high turbulence activities are essential in improving the mixing and enhancing the fuel air reaction.

Figure 14 demonstrates the RMSA contours in the X-Z for the base configuration. We notice that the dome presence

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results in a decrease of the amplitude of oscillations close to the swirler exit. However, the turbulence decays downstream at a distance of 7R (it decays around 4R for configuration 1).

4.2.2 Radial velocity and turbulent fluctuating component:

The radial velocity profiles corresponding to the axial velocity profiles discussed previously are presented for configuration 1 and the base configuration in Figures 15 and 16 respectively.

Figure 15 shows that as the flow emanates from the flare, it suddenly expands and there is a tremendous increase in the radial velocity. However, the radial velocity diminishes in the vicinity of the combustor centerline. As the flow proceeds downstream, the radial velocity peaks starts moving away from the centerline. This is accompanied by a decrease in the velocity magnitude as the flow spreads over the cross section. At a downstream distance of 20mm the expanding jet reaches the combustor walls. The jet impinges and bounces off the wall at a distance of around 40mm as shown in Figure 15 (radial velocity profiles switch direction). Toward the end of the combustor, the radial velocity vanishes indicating that the flow becomes fully developed.

The base configuration radial velocity profiles, illustrated in Figure 16, show similar behavior to that of configuration 1. However, the radial velocity is slightly higher on the dome side with the higher expansion angle (right hand side). We also notice that there are almost no radial velocity humps in the region close to the swirler exit. In other words, the expanding jet reaches its maximum velocity at the expanding surface of the dome. This is a major difference between both configurations. Unlike configuration 1, at Z/R=8 the radial velocity doesn't vanish.

The radial velocity rms contours (RMSR) for configuration 1 and the base configuration are presented in Figures 17 and 18 respectively. The difference between the rms values of the axial and radial velocity components is significant in the vicinity of the swirler exit especially in the expanding jet region. The high anisotropy of turbulence in that region may be the reason that existing turbulence models fail to predict the swirling flow close to the swirler exit.

4.3 Vertical plane (Y-Z) measurements for both configurations:

4.3.1 Axial velocity and turbulent fluctuating component:

The axial velocity profiles for configuration 1 and the base configuration are presented in Figures 19 and 20 respectively.

It is interesting to find out that both configurations have very similar axial velocity profiles in the Y-Z plane which is attributed to the fact that both configurations have no restriction on the flow in the Y-Z plane. As a result of this similarity, the discussion in the current section is limited to configuration 1.

We notice that up to a distance of 4R the velocity inside the CRZ is still 10 m/s (similar to the X-Z results). It is attention-grabbing that the velocity magnitude is similar even though the chamber has a rectangular cross section. In other words, along the short side or the long side the velocity inside the CRZ is still the same (10 m/s). However, the jet velocity is relatively higher (comparing the Y-Z to the X-Z) as illustrated by the contours of axial velocity in Figure 21 which might be attributed to two factors. First, the chamber doesn't have a square cross section. It has a rectangular cross section with the short side in the Y-Z plane. In other words, the increase in velocity occurs due to the relative reduction in the forward flow area. Second, the presence of the splash plate lip in the Y-Z direction only. The CRZ maximum width in the Y-Z plane is around 3.2Rwhile the maximum width in the X-Z plane is around 3.8R and so the ratio is around 0.85. It is appealing that the chamber rectangular cross section also has a width to breadth ratio of 0.85 which highlights the strong influence of the confinement on the shape of the CRZ.

4.3.2 Tangential velocity and turbulent fluctuating component:

The tangential velocity profiles for configuration 1 and the base configuration are presented in Figures 22 and 23 respectively. Since both configurations show similar behavior, we will focus only on configuration 1.

The swirl cup has two radial swirlers with counter-rotating vanes arrangement. However, the primary swirler has a larger effective area than the secondary swirler. It is interesting to see that at 5mm the central region of the combustor has almost zero tangential velocity as shown in Figure 22. This is an indication of the fact that the mixing occurred between the primary and secondary flow inside the venturi and before the flow reached the 5mm downstream distance. However, we can still see that there is a small portion (from y/R=0.4 to 0.8) of the curve that has a counter rotation to the rest of the flow. The tangential velocity profile at Y/R=0.2 also shows a typical Rankin vortex behavior (a forced or body vortex followed by a free vortex).

As we proceed downstream, we can see that the flow field is entirely rotating in the counter clockwise direction as shown in Figure 22. We can see that up to Y/R=0.8 the tangential velocity profiles are not smooth. This is expected because of the highly 3D unsteady complex flow close to the swirler exit. This is the region that CFD fails to capture. The tangential velocity profiles continue to show a Rankin vortex behavior up to a downstream distance of Y/R=1.6. However, as we proceed downstream, the swirling flow becomes fully developed and the tangential velocity profiles become smooth and follow a forced vortex behavior.

Figure 24 shows the tangential velocity contours and the corresponding rms values (RMST) for configuration 1. The swirling flow is fairly symmetric around the combustor centerline. We can see that in the vicinity of the combustor centerline, no swirl is present. High swirling velocities are present in the jet region and extend downstream up to a distance of 1R. High turbulence activities occur in the jet region

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and around the edge of the CRZ as depicted in Figure 24. In addition, we still notice high anisotropy in the turbulence stresses (compare the axial and tangential rms).

4.4 Confinement effect on the flow field:

The axial and radial velocity contours of the base configuration installed in the SAC sector are shown in Figure 25. The CRZ shape is strongly influenced by the confinement as we notice that the CRZ height reduces from 8R (rectangular chamber) to 4.8R (SAC sector). The maximum width is almost the same. However, the CRZ is much fatter. The variation of the size of the CRZ will have a significant influence on the reacting flow characteristics. Despite the fact that the size of the CRZ is directly related to the amount of mass being recirculated back and consequently to the heat release rate, reacting flow measurements are needed to verify such statements. The axial velocity inside the CRZ in both confinements is around 10 m/s up to 3R. The expanding swirling jet radial velocity is similar in both confinements up to 1R.

Usually real combustion chambers are designed with a contraction toward the combustor end such that the combustor exit will fit the turbine nozzle ring. The effect of the contraction on the flow field is apparent from the streamlines as illustrated in Figure 25. The flow field is tilted in the same direction of the inclined wall and that effect extends further away from the inclined way. It almost extends up to the other chamber wall as shown in Figure 25. Radial velocity is high at the expanding jet region. It diminishes in all other regions but the area affected by the contracting combustor wall.

The axial and radial rms contours for the same configuration are illustrated in Figure 26. The highest turbulence activities are limited to the expanding jet region. It extends downstream up to a distance of 1.5R. As we proceed downstream, the rms value drops by about 50% in the vicinity of the CRZ edge.

The rms values of the velocities for the base configuration are different with different confinements. Figures 26 and 14 show noticeable differences in the rms contours in the expanding jet region, that fact emphasizes the importance of testing the hardware in geometries as close as possible to the real geometries.

5 CONCLUSIONS

The expanding annular swirling jet region emanating from the flare is shown to be a region of high amplitude oscillations. Thus, such a region becomes very sensitive and even a short lip is shown to have a considerable effect on the flow field.

The dome damps the velocity fluctuations in the expanding jet region. Hence, the sensitivity of the flow field to generation of periodic oscillations is expected to reduce significantly and therefore, more stable combustion could be achieved.

Oscillations in the axial, radial and tangential velocities are shown to have different magnitudes. This is noticeable in the annular expanding jet region.

The confinement and the method of admission of the swirling flow to the combustion chamber are shown to have significant effects on the shape of the CRZ. The SAC sector has similar dimensions to the rectangular chamber. However, the contraction toward the exit results in a reduction of the height of the CRZ by 40% even though the same swirl cup is used in both cases. Also, a rectangular chamber with width to breadth ratio of 0.85 resulted in a CRZ with approximately the same diameter ratio.

The contraction in the combustion chamber results in a flow field that is tilted in the same direction of the wall.

The presence of lips, combustion dome, asymmetry in the dome expansion angle and the shape of the confinement all result in an asymmetric CRZ which questions the CFD codes that assume symmetric CRZ in the simulation process.

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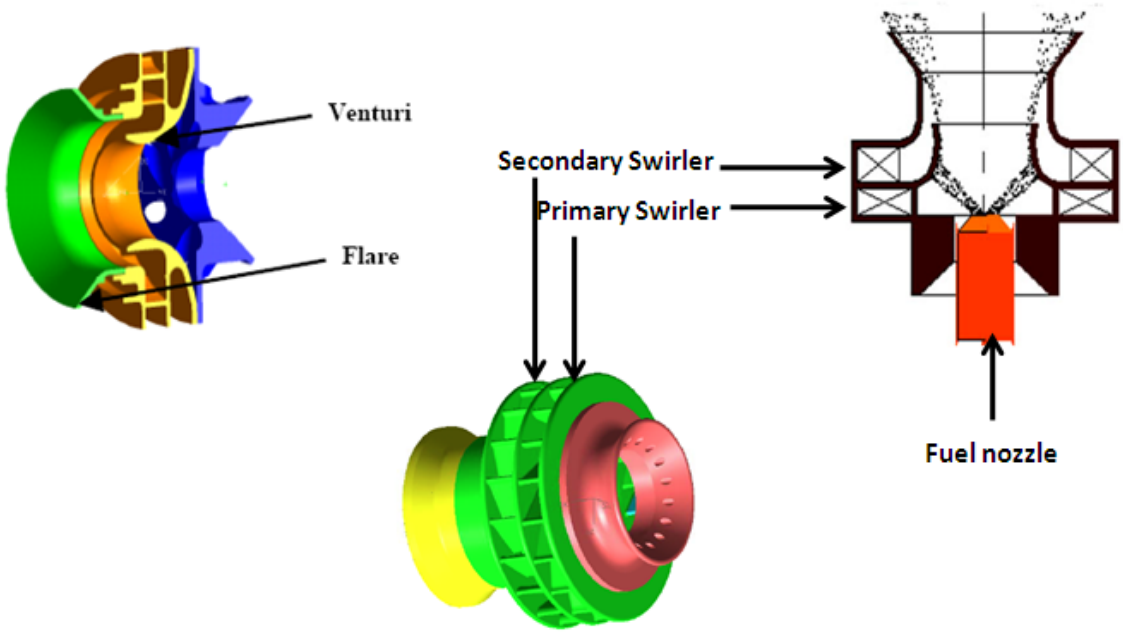

Figure 1. Typical swirl cup arrangement with counter rotating radial swirlers.

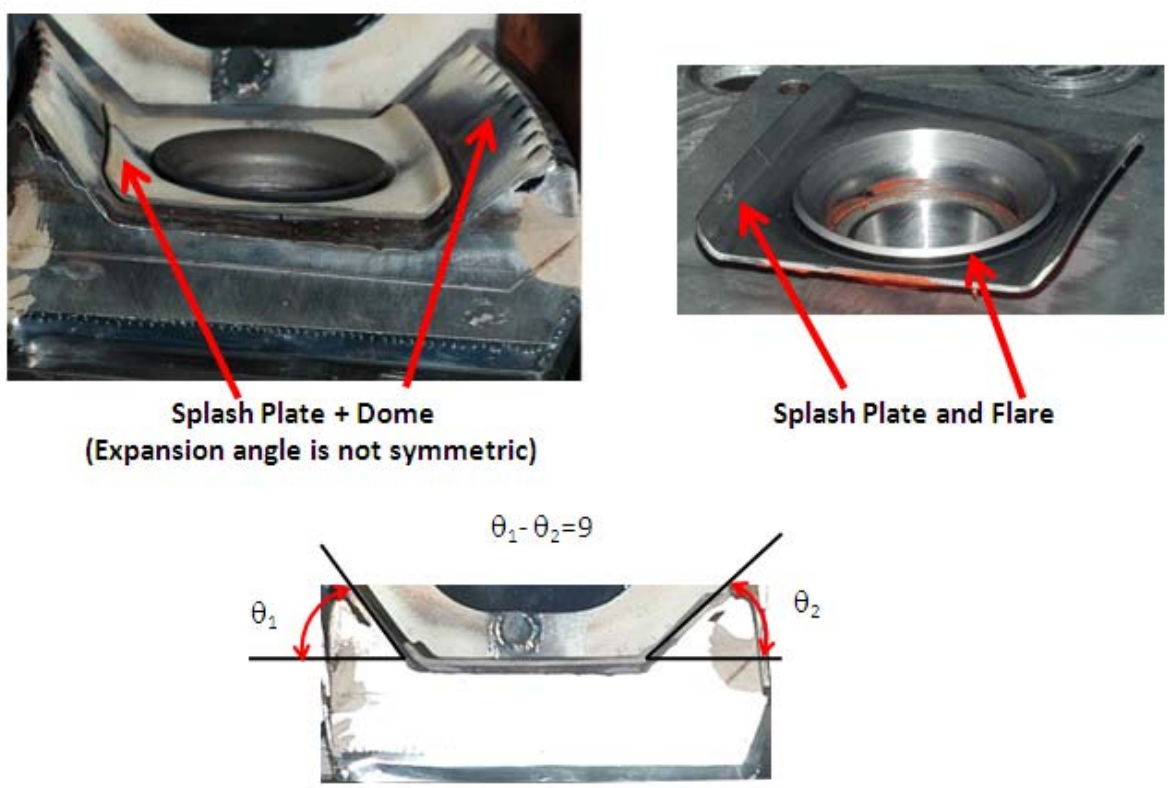

Figure 2. Configurations of the swirl cup under investigation.

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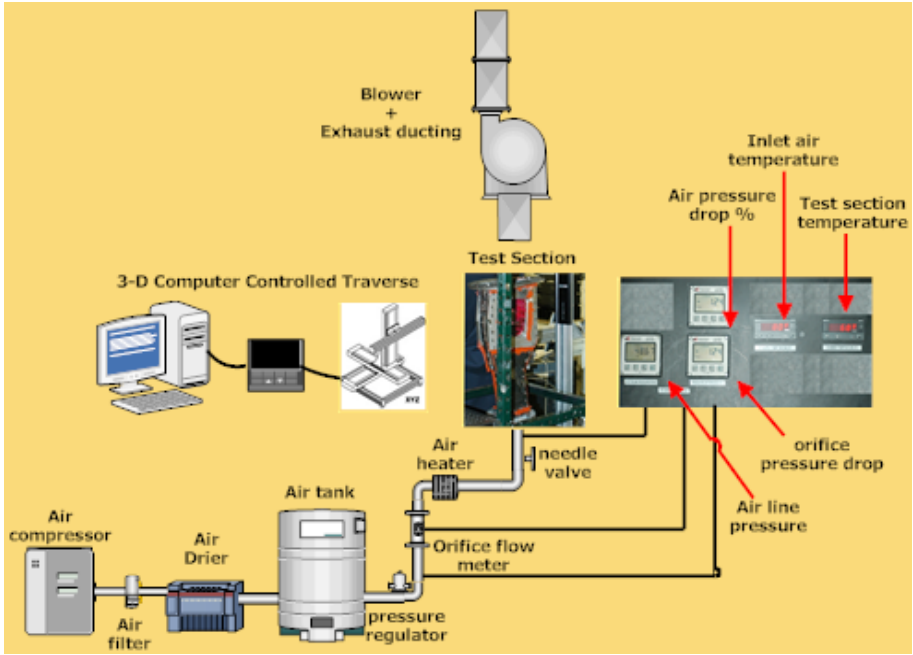

Figure 3. Schematic of the experimental facility to study the dump combustor aerodynamics

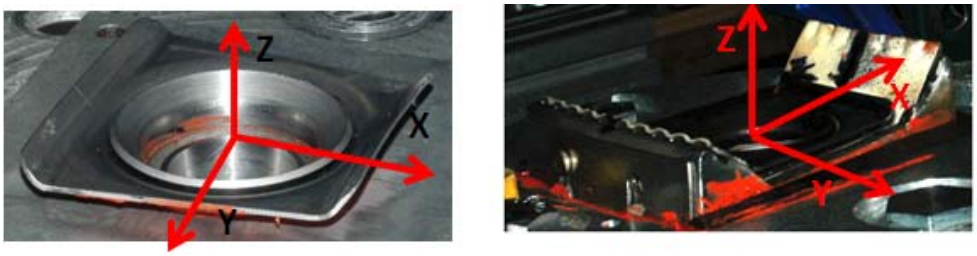

Figure 4. Measurements coordinate system for both configurations

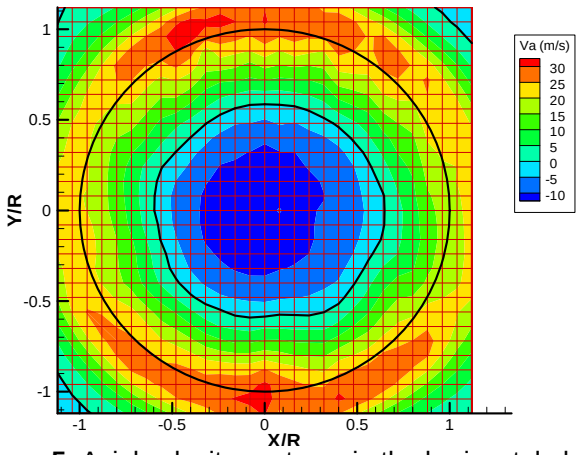

Figure 5. Axial velocity contours in the horizontal plane (configuration 1)

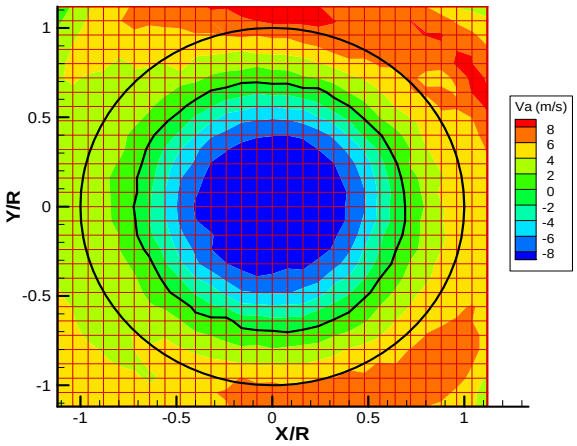

Figure 6. Axial velocity contours in the horizontal plane (base configuration)

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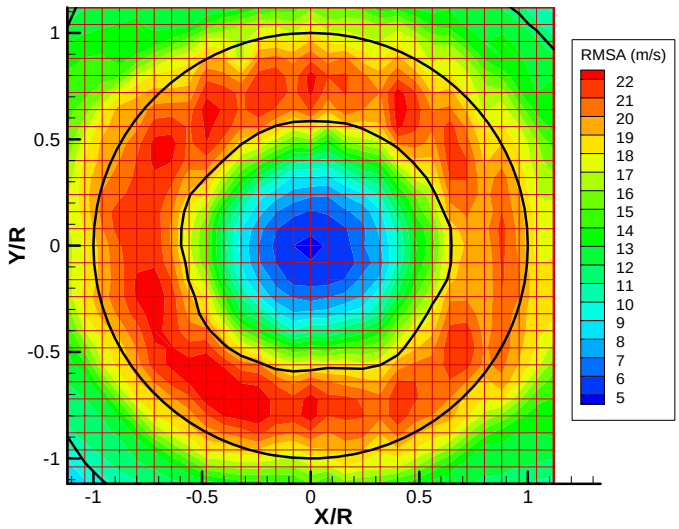

Figure 7. Contours of the root mean square of the axial velocity in the horizontal plane (configuration 1).

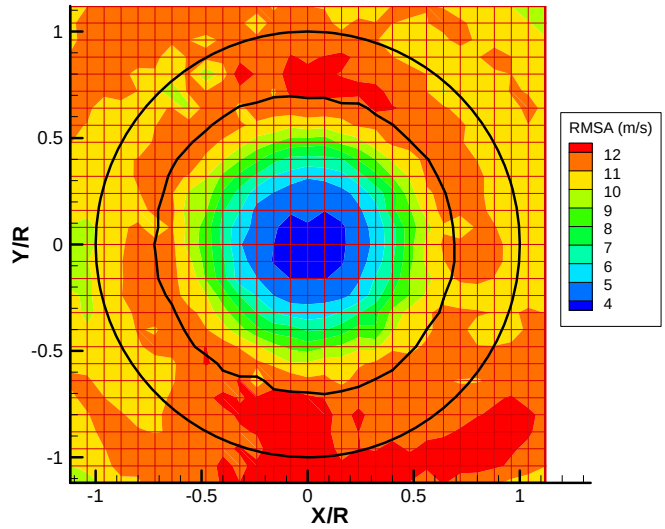

Figure 8. Contours of the root mean square of the axial velocity in the horizontal plane (base configuration).

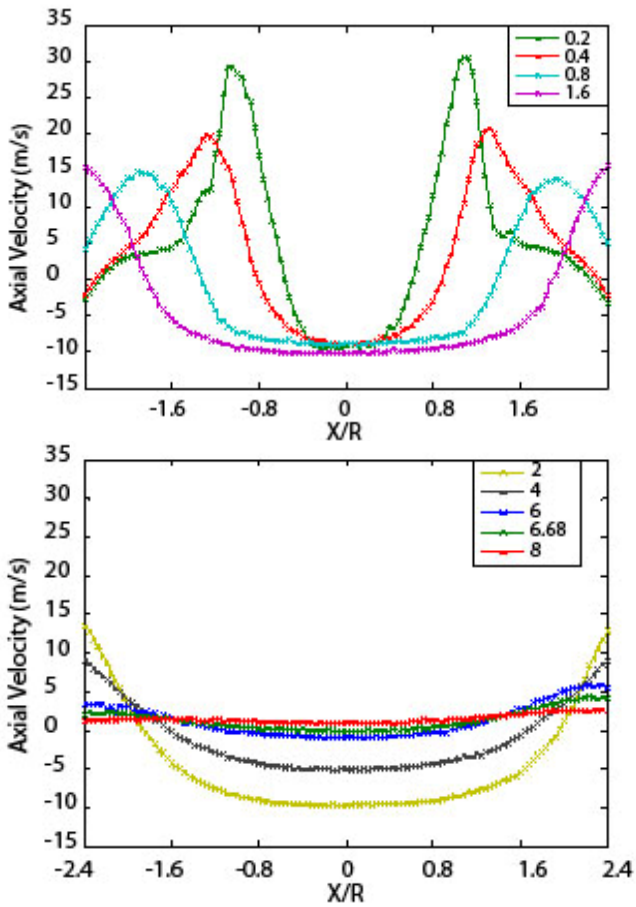

Figure 9. Axial velocity profiles in the x-z plane (Configuration 1)

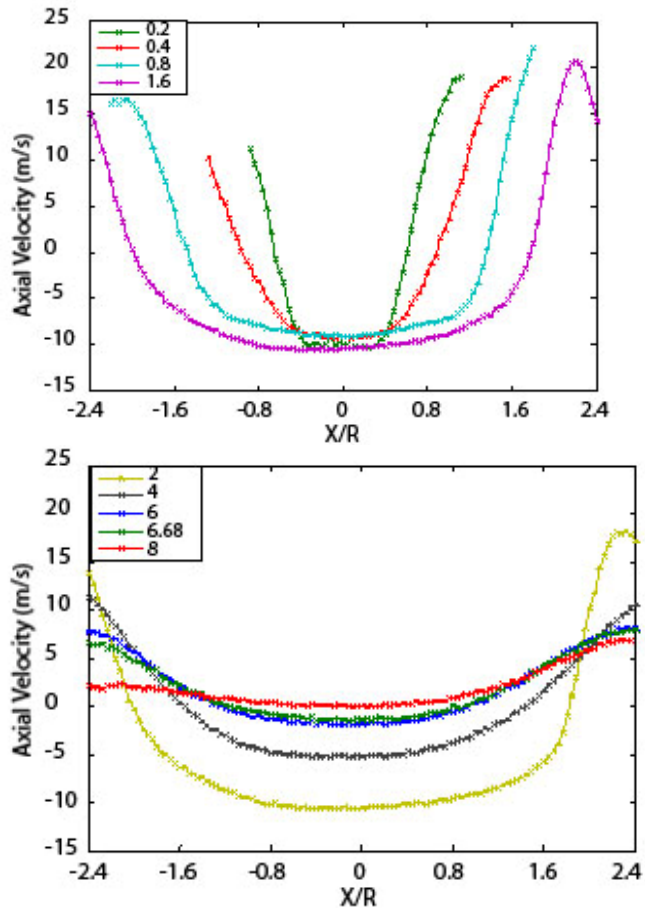

Figure 10. Axial velocity profiles in the x-z plane (base configuration)

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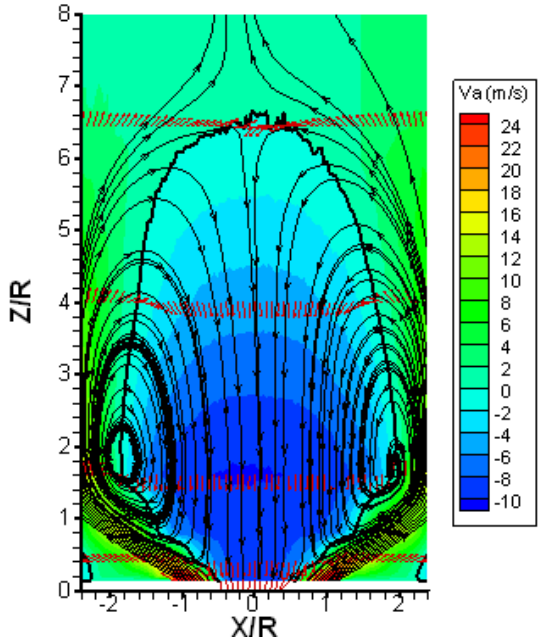

Figure 11. Axial velocity contours in the x-z plane (configuration 1)

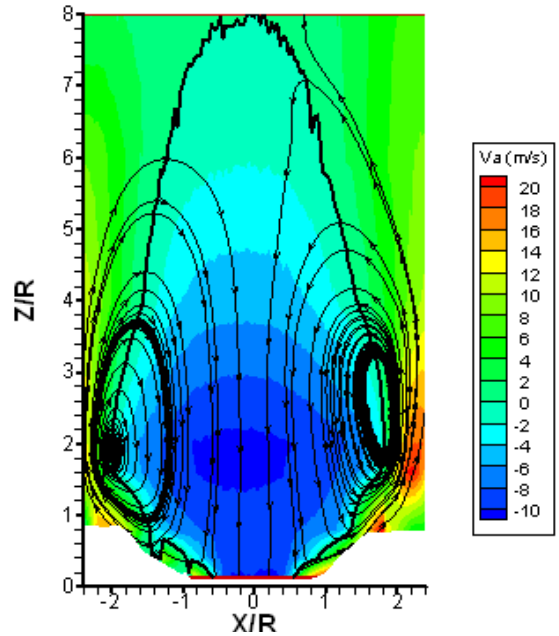

Figure 12. Axial velocity contours in the x-z plane (base configuration)

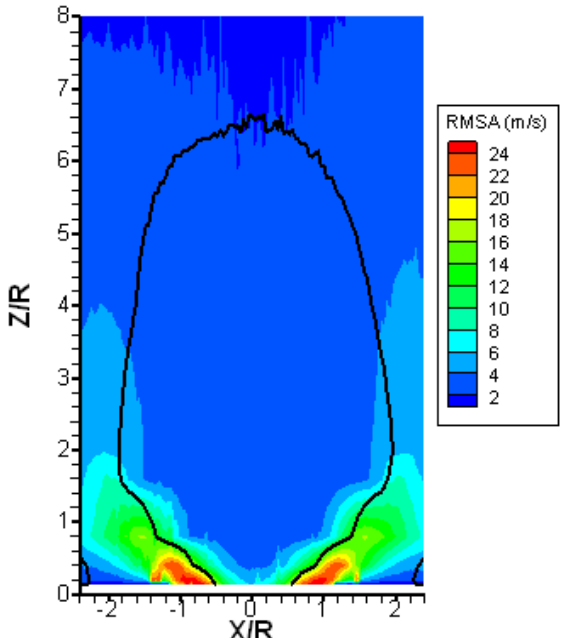

Figure 13. RMSA contours in the x-z plane (configuration 1)

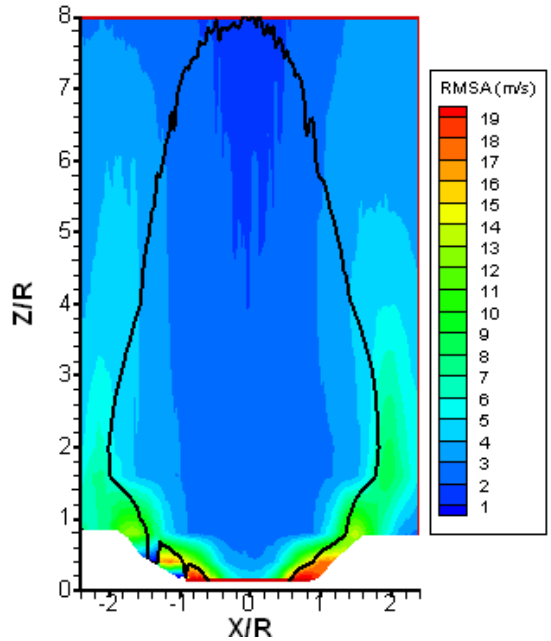

Figure 14. RMSA contours in the x-z plane (base configuration)

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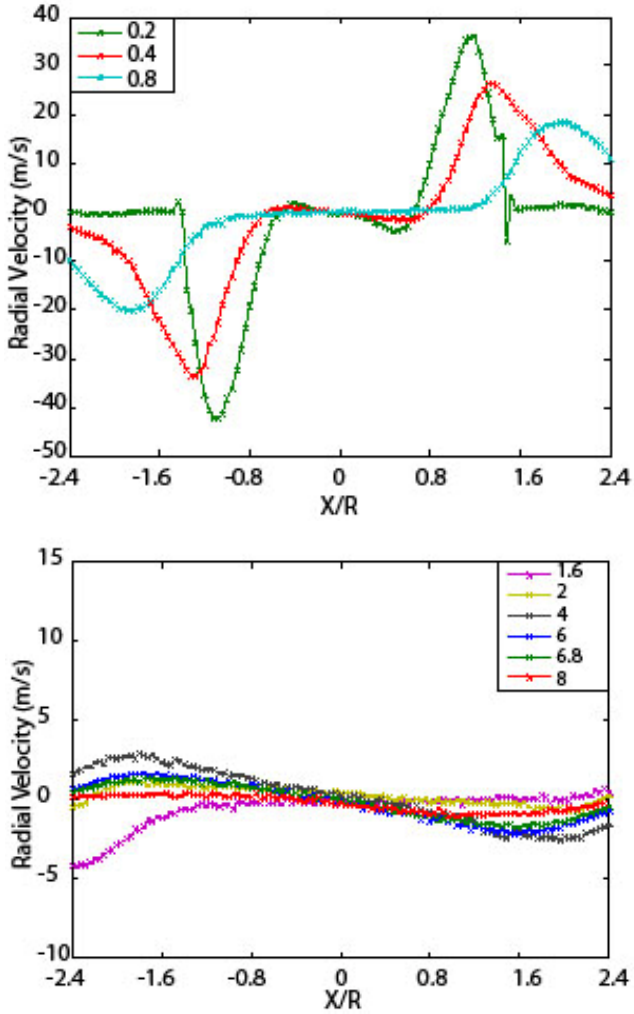

Figure 15. Radial velocity profiles in the x-z plane (configuration 1)

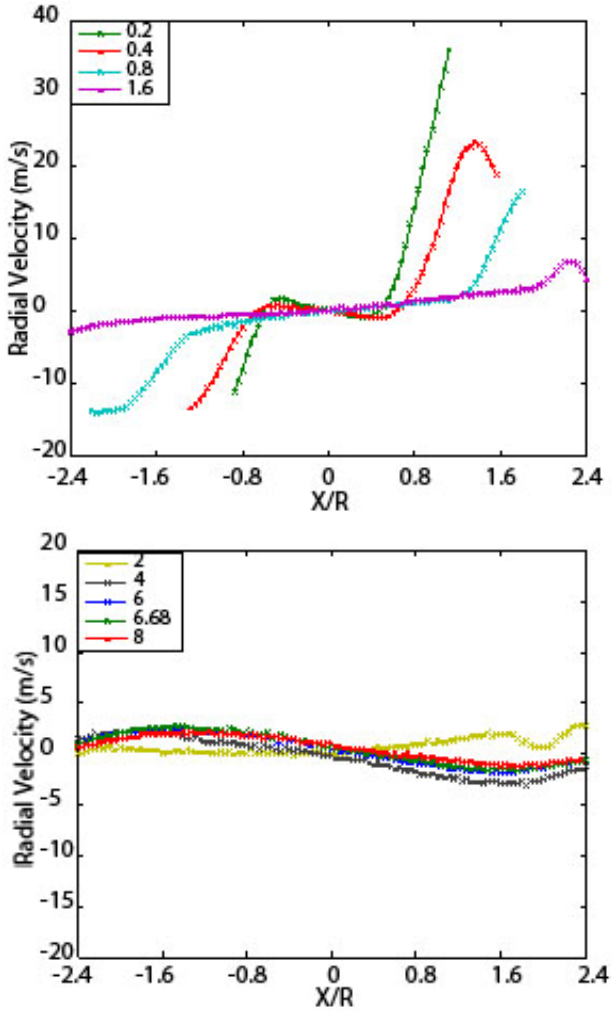

Figure 16. radial velocity profiles in the x-z plane (base configuration)

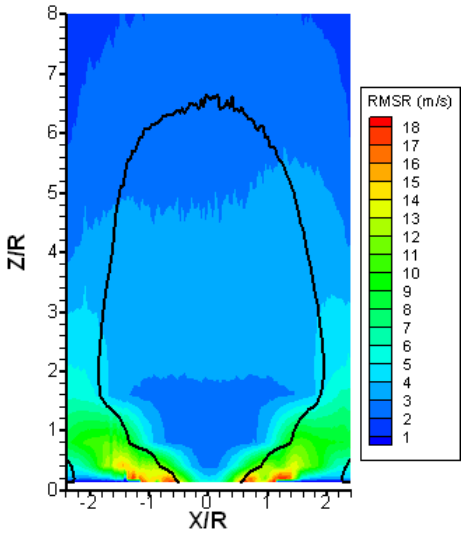

Figure 17. RMSR contours in the y-z plane (configuration 1)

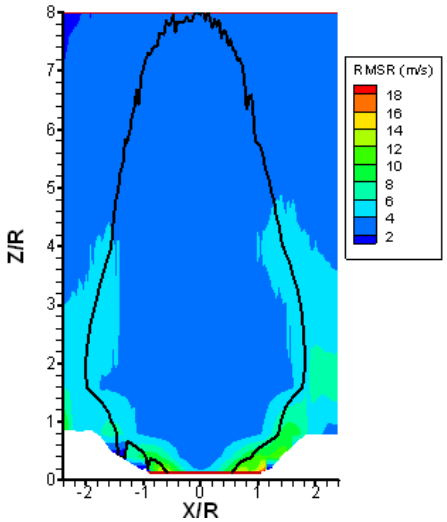

Figure 18. RMSR contours in the y-z plane (base configuration)

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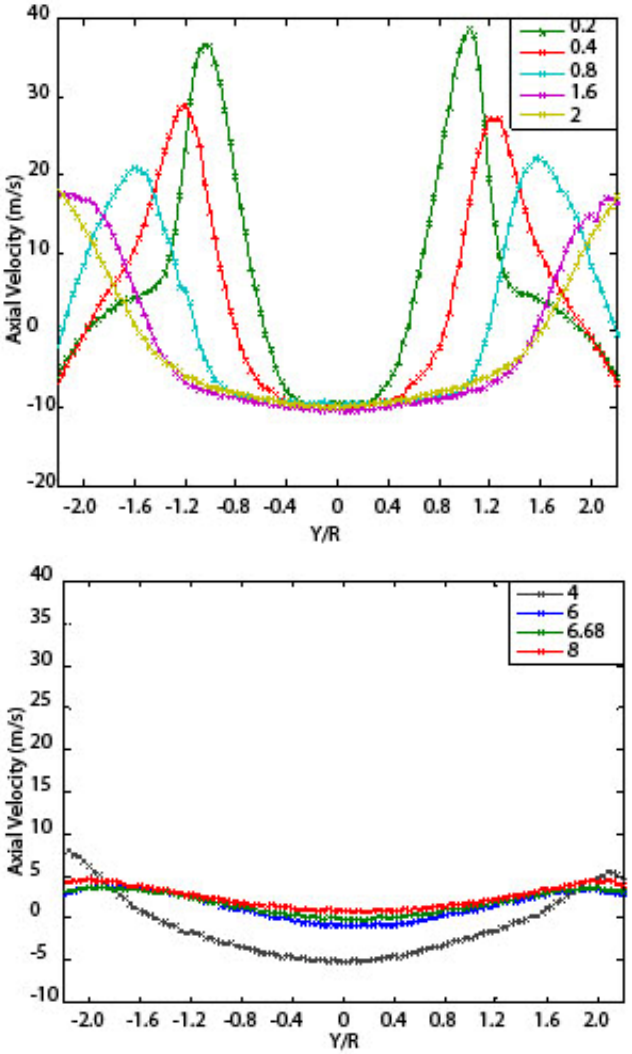

Figure 19. Axial velocity profiles in the y-z plane (configuration 1)

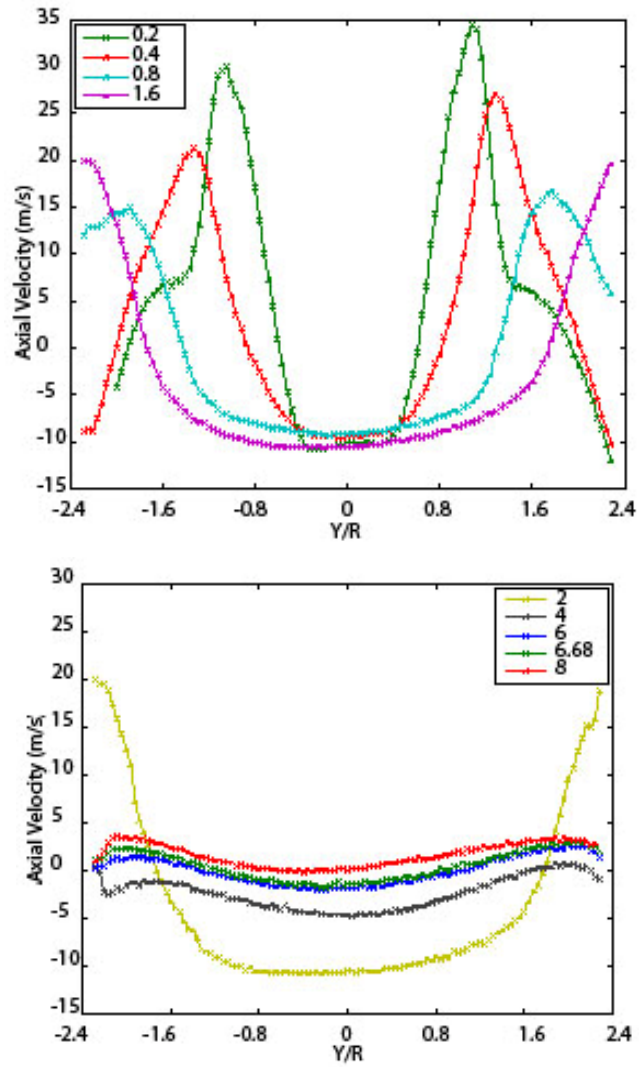

Figure 20. Axial velocity profiles in the y-z plane (base configuration)

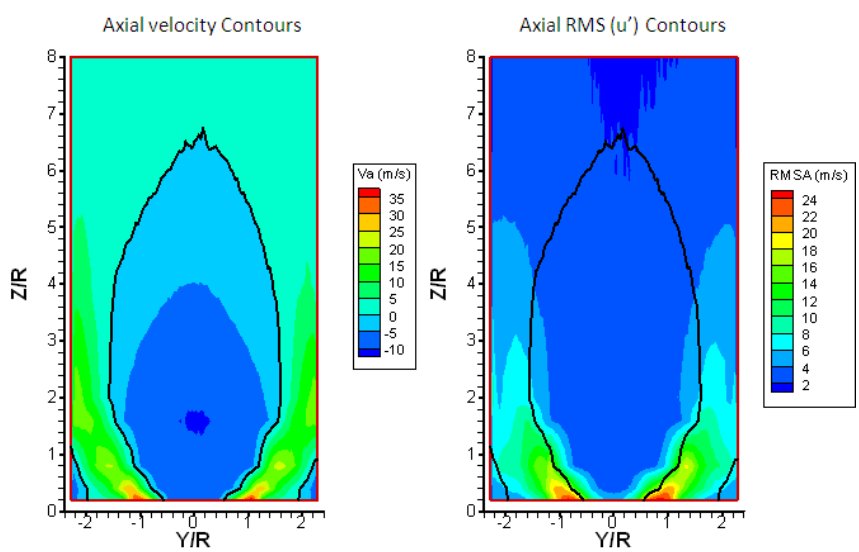

Figure 21. Axial velocity and RMSA contours in the x-z plane (Configuration 1)

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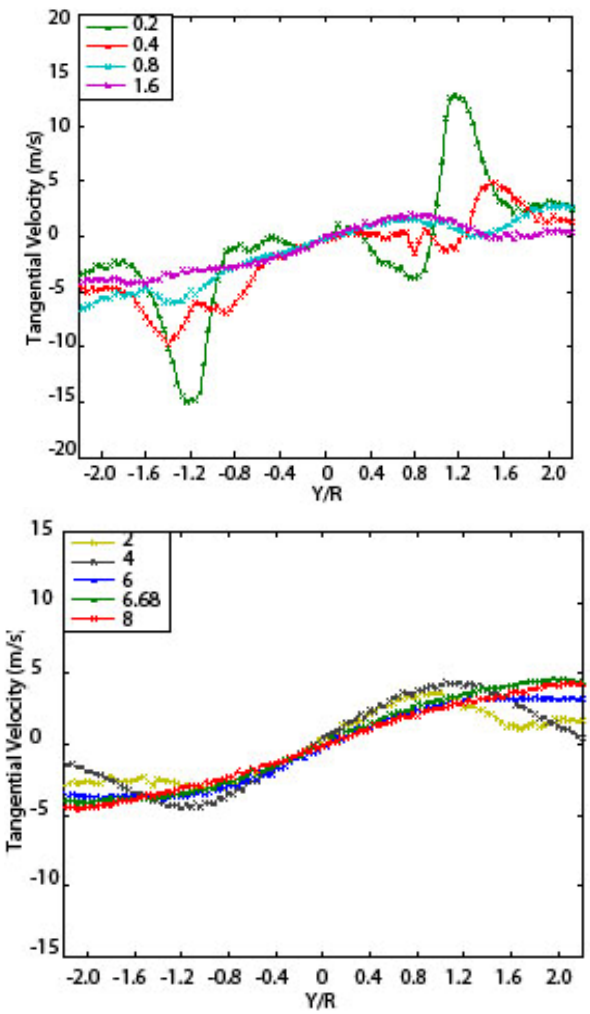

Figure 22. Tangential velocity profiles in the y-z plane (configuration 1)

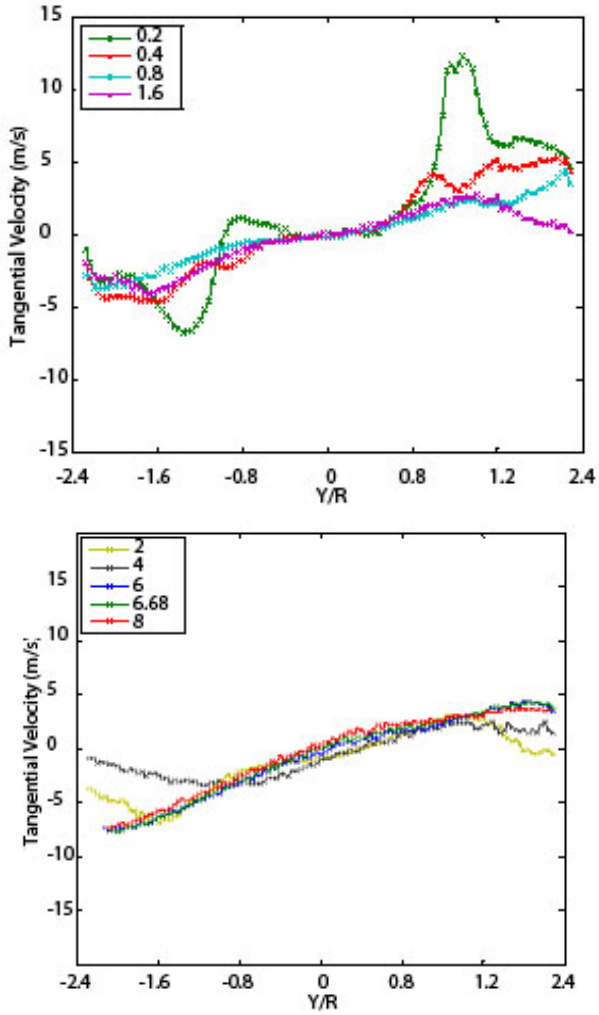

Figure 23. Tangential velocity profiles in the y-z plane (Base configuration)

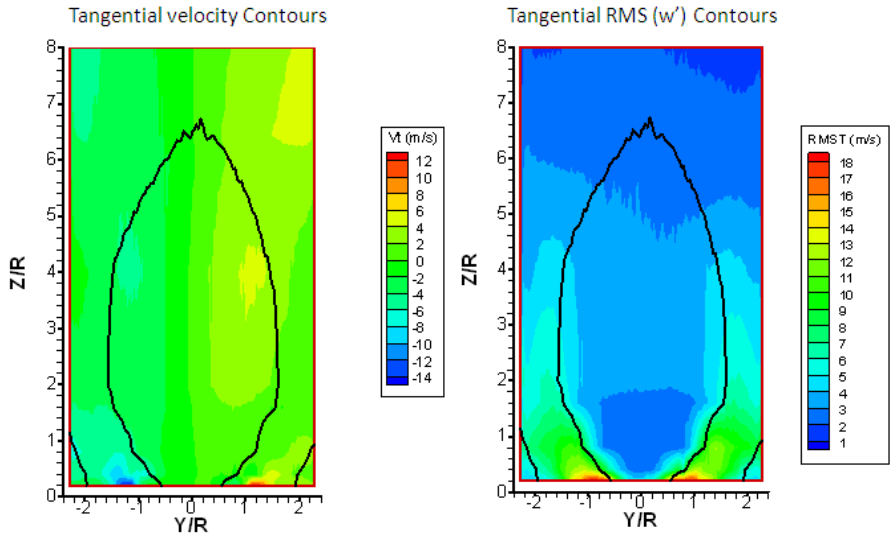

Figure 24. Tangential velocity and RMST contours in the y-z plane (configuration 1)

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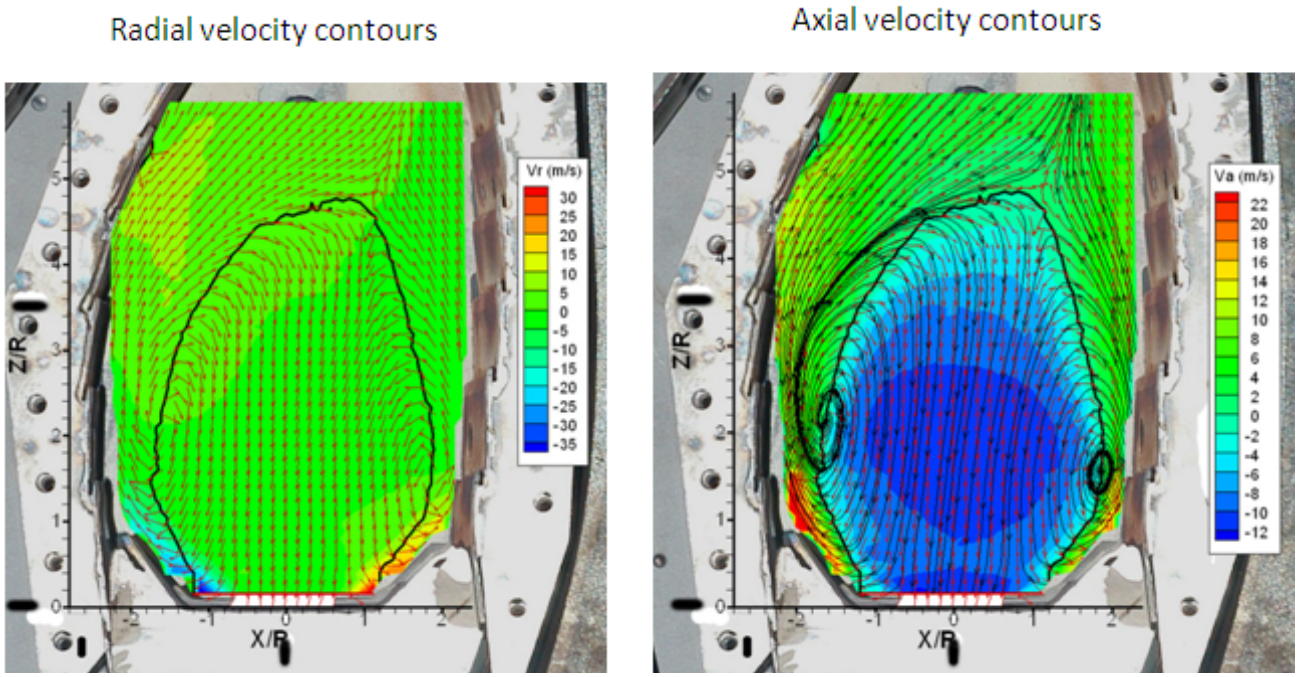

Figure 25. Axial and radial velocity contours of the base configuration installed in the SAC sector.

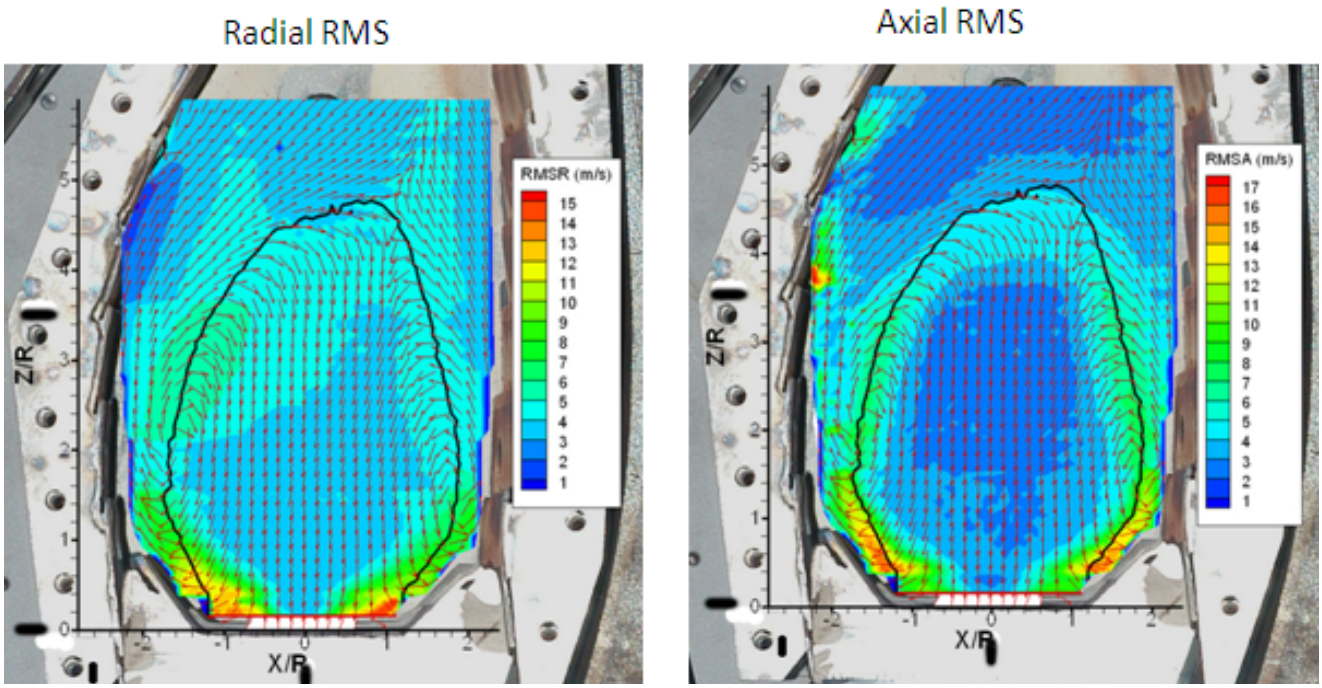

Figure 26. Axial and radial rms contours of the base configuration installed in the SAC sector.

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